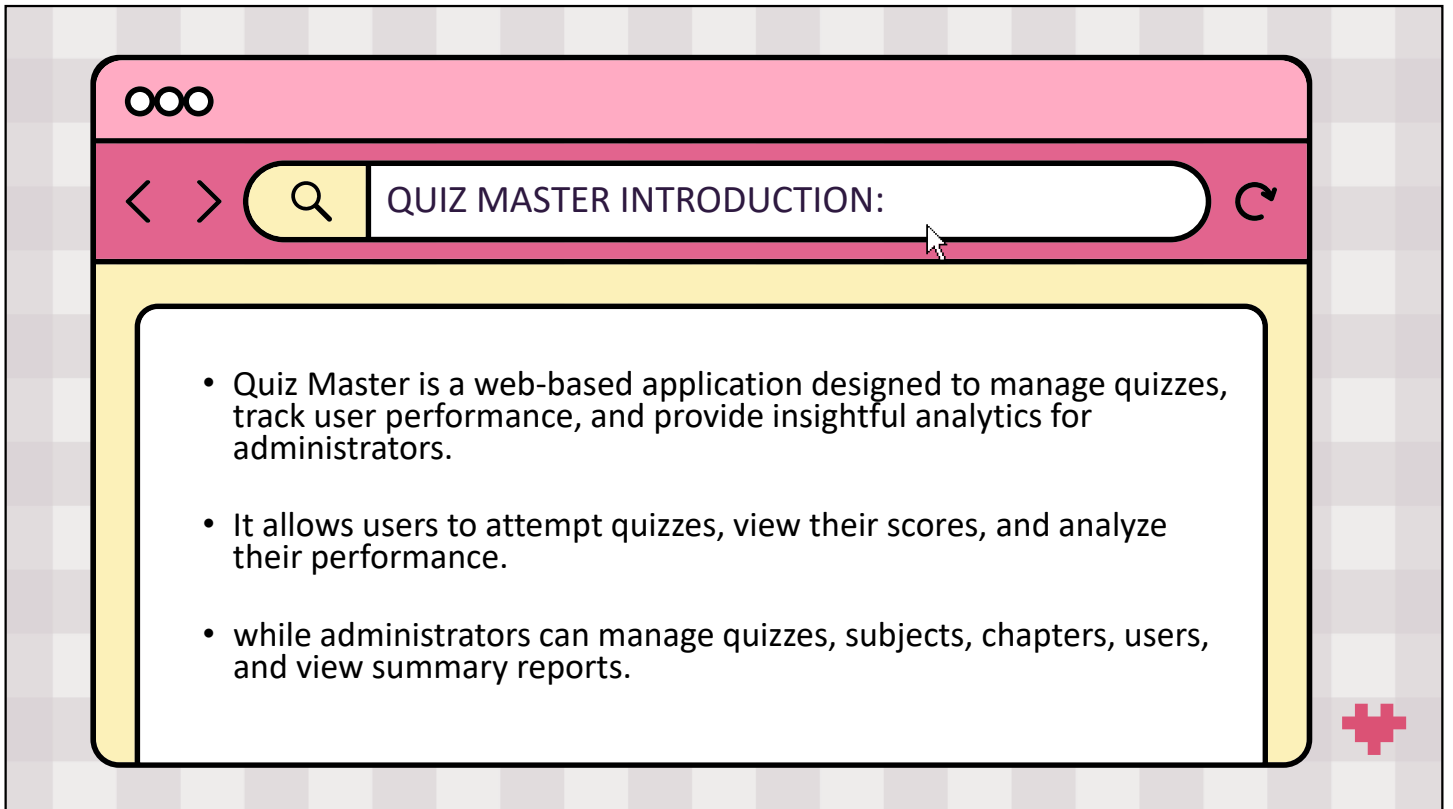
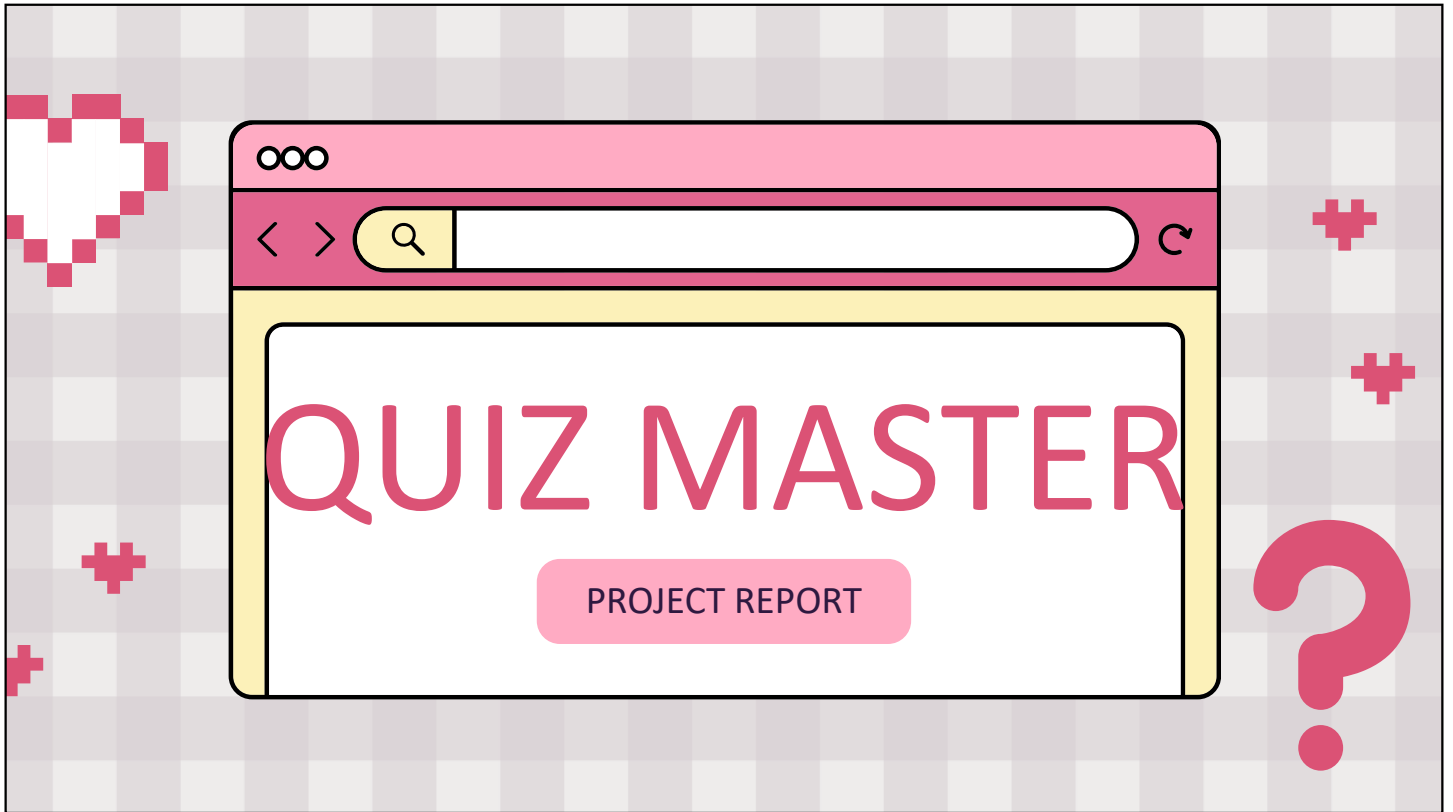




TECHNOLOGIES

BACKEND

FLASK: A PYTHON-BASED WEB FRAMEWORK FOR HANDLING APPLICATION LOGIC AND ROUTING.

SQLITE: A LIGHTWEIGHT RELATIONAL DATABASE FOR DATA STORAGE AND MANAGEMENT.

FLASK-SQLALCHEMY: ADDS SQLALCHEMY SUPPORT FOR DATABASE MANAGEMENT IN FLASK.

FRONTEND

HTML5: FOR STRUCTURING THE WEB PAGES.

CSS3: FOR STYLING AND RESPONSIVE DESIGN.

BOOTSTRAP: FOR A USER-FRIENDLY, RESPONSIVE, AND CONSISTENT INTERFACE.

JAVASCRIPT: FOR INTERACTIVE ELEMENTS AND DYNAMIC CONTENT LOADING

LIBRARIES AND TOOLS

JINJA2: FOR SERVER-SIDE TEMPLATING IN FLASK.

REQUESTS: A LIBRARY FOR MAKING HTTP REQUESTS IN PYTHON

MATPLOTLIB: LIBRARY FOR CREATING INTERACTIVE CHARTS

SERVER

WERKZEUG: FOR DEVELOPMENT SERVER AND REQUEST HANDLING IN FLASK.

DATABASE SCHEMA



THIS DATABASE SCHEMA OF QUIZ MASTER APPLICATION, WITH MODELS REPRESENTING USERS, SUBJECTS, CHAPTERS, QUESTIONS, QUIZZES, AND SCORES. HERE'S A BRIEF OVERVIEW OF THE MODELS AND THEIR RELATIONSHIPS:

- **USER:** STORES USER INFORMATION, INCLUDING CREDENTIALS AND PERSONAL DETAILS. IT HAS A ONE-TO-MANY RELATIONSHIP WITH SCORE, ALLOWING USERS TO HAVE MULTIPLE QUIZ SCORES.
- **SUBJECT:** Represents different subjects or topics for quizzes. It has A ONE-TO-MANY RELATIONSHIP WITH CHAPTER, ENABLING MULTIPLE CHAPTERS PER SUBJECT.
- **CHAPTER:** SUBDIVISIONS OF SUBJECTS. IT HAS ONE-TO-MANY RELATIONSHIPS WITH BOTH QUESTION AND QUIZ, ALLOWING CHAPTERS TO CONTAIN MULTIPLE QUESTIONS AND QUIZZES.
- **QUESTION:** INDIVIDUAL QUIZ QUESTIONS WITH OPTIONS AND CORRECT ANSWERS. IT HAS MANY-TO-ONE RELATIONSHIPS WITH CHAPTER AND QUIZ, ASSOCIATING EACH QUESTION WITH A SPECIFIC CHAPTER AND OPTIONALLY WITH A QUIZ.

DATABASE SCHEMA



- **QUIZ:** REPRESENTS A COMPLETE QUIZ. IT HAS MANY-TO-ONE RELATIONSHIP WITH CHAPTER, MANY-TO-MANY RELATIONSHIP WITH QUESTION (THROUGH THE QUIZ_ID in Question model), and ONE-TO-MANY RELATIONSHIP WITH SCORE.
- **SCORE:** TRACKS USER PERFORMANCE ON QUIZZES. IT HAS MANY-TO-ONE RELATIONSHIPS WITH BOTH USER AND QUIZ, LINKING EACH SCORE TO A SPECIFIC USER AND QUIZ.

